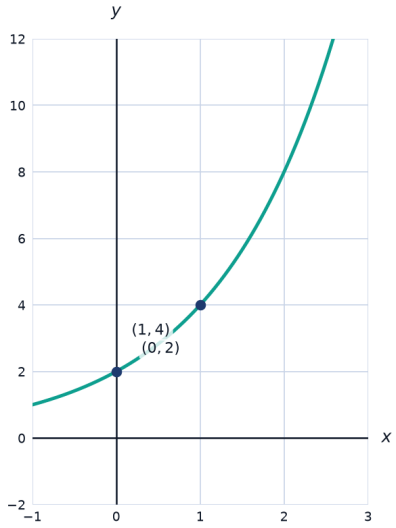
Exponentials, Radicals, Polynomials & Rational Expressions

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Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

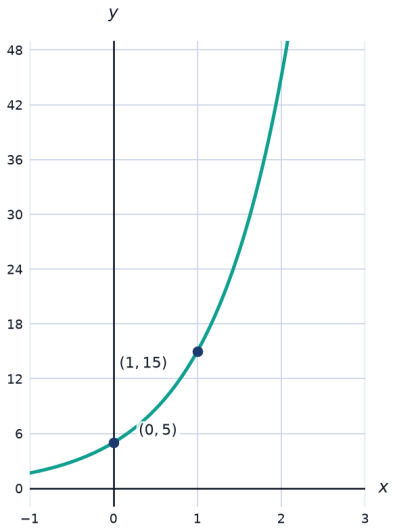
1 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the botanical conservatory is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



2 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

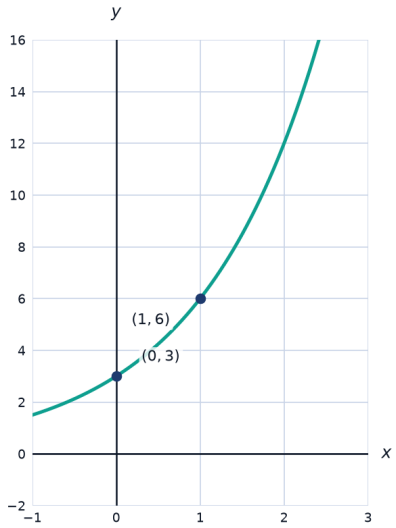
The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the regional bakery is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



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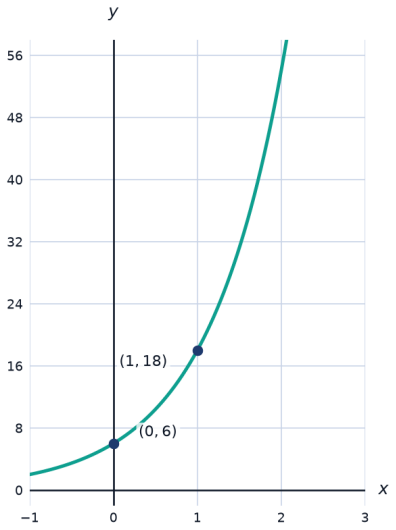
3 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the science museum is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



4 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

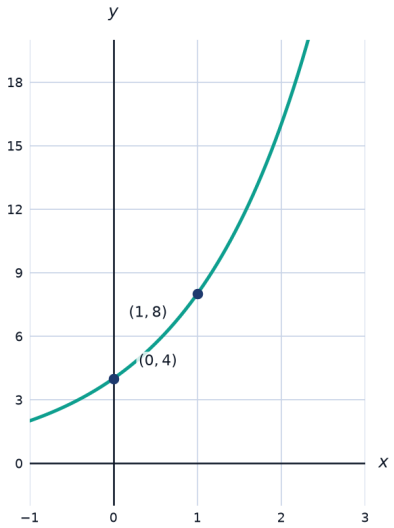
The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the ceramic studio is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



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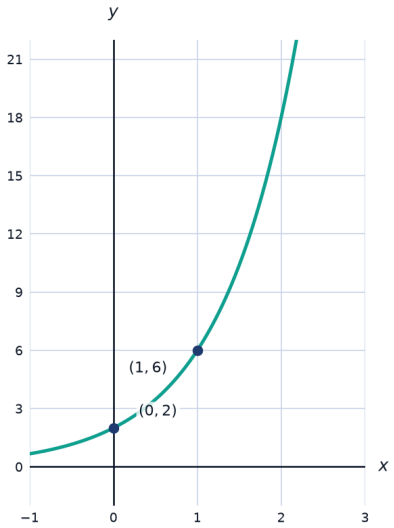
5 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the public transit depot is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



6 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

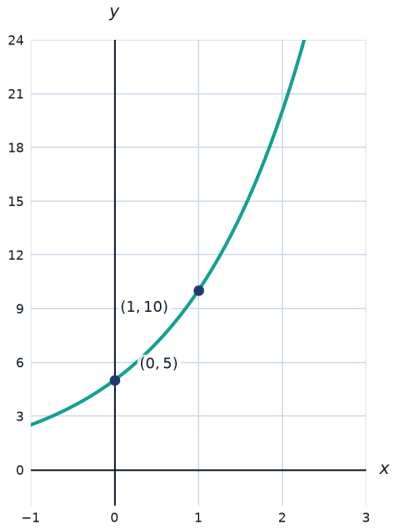
The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the art gallery is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



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7 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the university lecture hall is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $3^x = 81$, what is the value of x?

9 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the sports complex, in a calculation for the geology lab (case 40), when $(x + 4)(x + 1)$ is expanded, what is the coefficient of x?

10 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the astronomical society, simplify $\frac{x^2 + 11x + 24}{(x + 3)}$, then evaluate the result at $x = 1$.

11 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the film archive, in a calculation for the community theater (case 52), when $(x - 1)(x + 7)$ is expanded, what is the coefficient of x?

12 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the mountain observatory, simplify $\frac{x^2 - 6x - 16}{(x + 2)}$, then evaluate the result at $x = -6$.

13 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the recycling center, in a calculation for the university lecture hall (case 60), solve for x: $\sqrt{x + 8} = 4$.

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14 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), add: $(8x^2 - 4x + 3) + (2x^2 - 7x - 8)$

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x > 0$, the expression $x \cdot \sqrt[2]{x^3}$ at the botanical conservatory can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

16 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the municipal library, for $x \neq -3$, the rational expression $\frac{3x+15}{x+3}$ is equivalent to $3 + \frac{k}{x+3}$. What is the value of the constant k?

17 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A culture at the community theater begins with 300 cells and doubles every 5 hours according to $P(t) = 300(2)^{\frac{t}{5}}$. How many cells are in the culture after 20 hours?

18 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x > 0$, the expression $x \cdot \sqrt[2]{x^6}$ at the regional bakery can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

19 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the coastal aquarium, for $x \neq -6$, the rational expression $\frac{6x+45}{x+6}$ is equivalent to $6 + \frac{k}{x+6}$. What is the value of the constant k?

20 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

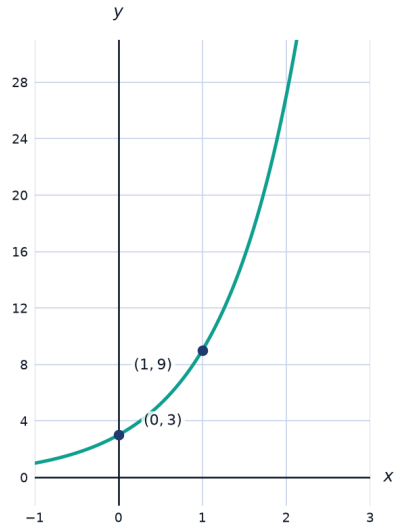
A culture at the mountain observatory begins with 450 cells and doubles every 4 hours according to $P(t) = 450(2)^{\frac{t}{4}}$. How many cells are in the culture after 16 hours?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the municipal library is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 27

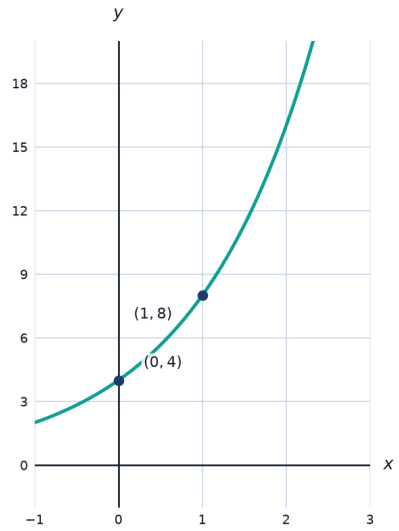
b. 21

c. 84

d. 81

22 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the community theater is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 36

b. 35

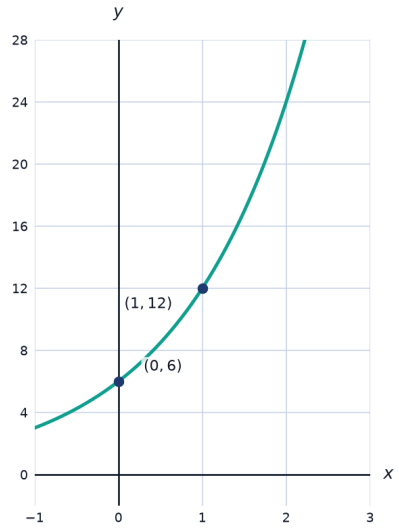
c. 32

d. 16

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23 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the coastal aquarium is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 48

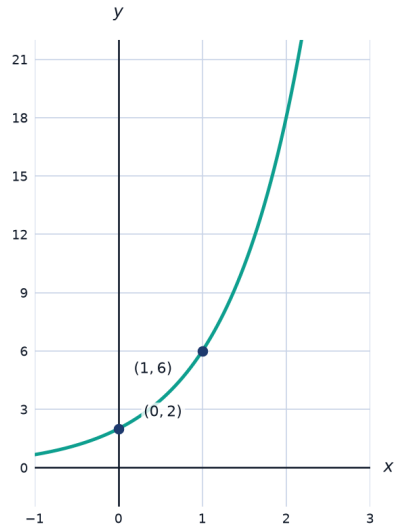
b. 24

c. 54

d. 51

24 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the mountain observatory is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 18

b. 14

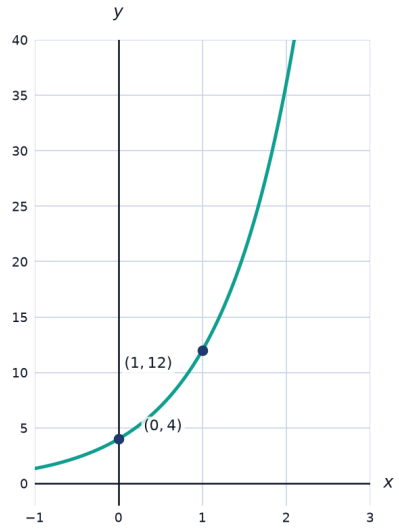
c. 56

d. 54

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25 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the urban bike-share station is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 112

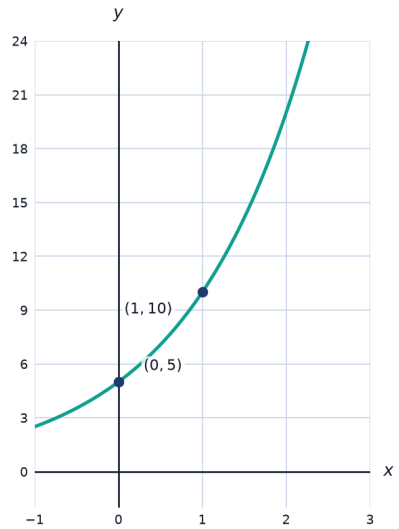
b. 108

c. 36

d. 28

26 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the solar microgrid facility is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 40

b. 20

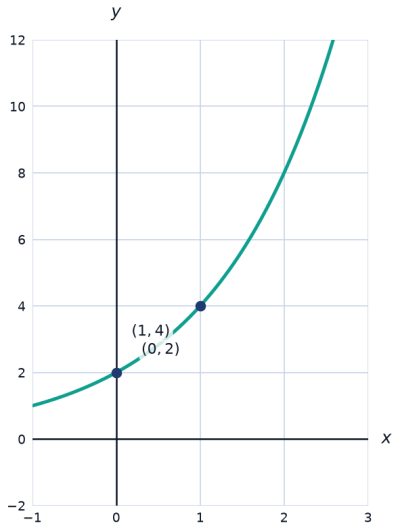
c. 45

d. 43

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27 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the agricultural cooperative is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 18

b. 19

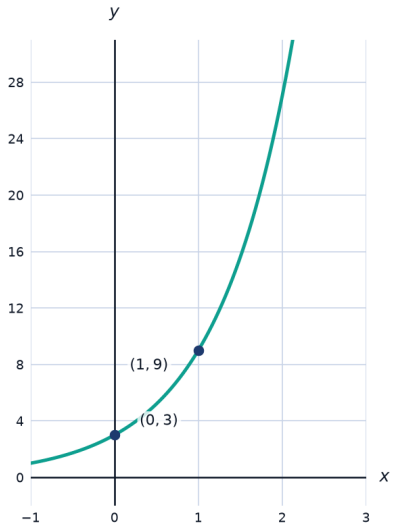
c. 16

d. 8

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28 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the publishing house is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?



a. 84

b. 81

c. 27

d. 21

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the robotics club, in a calculation for the orchestra hall (case 12), simplify: $(b^8)^2$.

a. b^{16}

b. b^{18}

c. b^{64}

d. b^{10}

30 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the culinary institute, in a calculation for the charter ferry service (case 13), subtract: $(2x^2 + 4x + 7) - (6x^2 + x + 3)$

a. $-4x^2 + 3x + 4$

b. $-2x^2 + 3x + 4$

c. $-4x^2 + 5x + 4$

d. $-4x^2 + 3x + 6$

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31 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), simplify: $(a^5)^3$.

a. a^{15}

b. a^8

c. a^9

d. a^{17}

32 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the sports complex, in a calculation for the geology lab (case 15), subtract: $(6x^2 + 5x - 3) - (2x^2 + 2x - 5)$

a. $6x^2 + 3x + 2$

b. $4x^2 + 3x + 4$

c. $4x^2 + 5x + 2$

d. $4x^2 + 3x + 2$

33 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the maple syrup farm, in a calculation for the honeybee apiary (case 16), add: $(8x^2 - 3x + 5) + (2x^2 - x - 3)$

a. $12x^2 - 4x + 2$

b. $10x^2 - 4x + 4$

c. $10x^2 - 2x + 2$

d. $10x^2 - 4x + 2$

34 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), simplify: $(y^8)^5$.

a. y^{14}

b. y^{42}

c. y^{13}

d. y^{40}

35 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), simplify: $(x^6)^9$.

a. x^{16}

b. x^{56}

c. x^{54}

d. x^{15}

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36 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), simplify: $y^5 \cdot y^7$.

a. y^2

b. y^{12}

c. $2y^{12}$

d. y^{35}

37 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), simplify: $b^7 \cdot b^4$.

a. b^{28}

b. b^3

c. $2b^{11}$

d. b^{11}

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the maple syrup farm, a radioactive isotope decays at a rate of 12% per day. If a sample initially has a mass of 80 grams, which function gives the mass $M(t)$ in grams remaining after t days?

a. $M(t) = 80(0.12)^t$

b. $M(t) = 80(0.88)^t$

c. $M(t) = 80(1.12)^t$

d. $M(t) = 80 - 0.12t$

39 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the planetarium, simplify $\left(\frac{4x^4}{y^6}\right)^{-1/2}$ for positive x and y.

a. $\frac{y^3}{2x^2}$

b. $\frac{2x^2}{y^3}$

c. $\frac{y^3}{4x^2}$

d. $\frac{2y^3}{x^2}$

40 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 68), if $x^3 - 4x^2 + 5x - 2 = (x-1)(x-2)Q(x)$, what is $Q(x)$?

a. $x - 1$

b. $x + 1$

c. $x - 2$

d. $x + 2$

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41 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the wind energy cooperative, in a calculation for the community orchard (case 69), solve for x: $\sqrt{3x+1}+4=9$.

a. 7

b. 8

c. 9

d. 10

42 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the marine research vessel, which of the following is equivalent to $\frac{2x}{x+3}+\frac{6}{x+3}$ for $x \neq -3$?

a. 2

b. $2x$

c. $\frac{2x+6}{(x+3)^2}$

d. $\frac{8x}{x+3}$

43 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x>0$, the expression $x\cdot\sqrt[2]{x^5}$ at the planetarium can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

a. 9

b. 7

c. 10

d. 5

44 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the geology lab, for $x \neq -6$, the rational expression $\frac{2x+18}{x+6}$ is equivalent to $2+\frac{k}{x+6}$. What is the value of the constant k?

a. 6

b. 18

c. 12

d. 8

45 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A culture at the regional airport shuttle begins with 450 cells and doubles every 4 hours according to $P(t)=450(2)^{\frac{t}{4}}$. How many cells are in the culture after 16 hours?

a. 3600

b. 7650

c. 7203

d. 7200

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